

Q.1 Multiple Choice Questions.

10

1. _____ is an acyclic diagram which has one node called its root whose in degree is 0 while other nodes have in degree 1.

A. Path.	B. Directed tree.
C. Edge.	D. Height.
2. The nodes which have the same parent are called _____.

A. Degree.	B. Leaf.
C. Height.	D. Siblings.
3. The node which is at the end and which does not have any child is called _____.

A. Branch.	B. Leaf.
C. Parent.	D. Root.
4. A graph G is said to be _____ if each vertex V_i is adjacent to every other V_j in G.

A. Multi graph	B. Complete graph
C. Multiple graph	D. Simple graph
5. A directed edge is known as _____.

A. Segment	B. Double edge
C. Arc	D. Arrow
6. A digraph is known as _____ graph.

A. Double graph	B. Multiple graph
C. Directed graph	D. Undirected graph
7. In selection sort algorithm, for ith pass of the sort, _____ comparisons are required.

A. $n + i$	B. $n - i$
C. $n + 1$	D. $n - 1$
8. The operation of merging K sorted tables into a single sorted table is called _____.

A. Selection sort	B. Insertion sort
C. K-way merging	D. Binary Merge
9. If there are no overflow records, the _____ is set equal to normal entry.

A. Normal entry	B. Overflow area
C. Prime entry	D. Master index
10. SIS stands for _____.

A. Symmetry Index Sequential	B. Simple Index Sequential
C. SCOPE Index Sequential	D. None of these

Q.2	Answer the following questions in short. (Any 10)	20
	1) Define: array	
	2) What is sparse matrix?	
	3) Define ancestor and subtree of a tree.	
	4) Draw the Binary Tree for $(A-B)+c*(E/F)$.	
	5) What is connected Graph?	
	6) Explain In degree, out degree and total degree of a Graph.	
	7) Difference between Searching and sorting.	
	8) List the applications of Searching.	
	9) List Searching and Sorting techniques.	
	10) Define : Record, Item	
	11) Define: Key, Sequence Key	
	12) Explain in brief overflow area.	
Q.3(A)	Define Tree. Explain 1-D array with declaration and initialization.	06
(B)	Explain applications of an array.	04
	OR	
Q.3(A)	Define Child Node. Explain 2-D array with declaration and initialization.	06
(B)	Explain applications of tree.	04
Q.4(A)	$(A/B+C)*(C+D*F)+(J/F)*(G/H)$. Draw the binary tree and find the pre order and post order of a binary tree.	10
	OR	
Q.4(A)	What do you mean by traversal of a tree? Explain the difference between Preorder and Post order with the help of illustration.	10
Q.5(A)	Write down the algorithm of Binary Search.	06
(B)	Define the all step of bubble sort as an ascending order of following data:	04
	55, 22, 77, 92, 12	
	OR	
Q.5(A)	Write down the algorithm of Sequential search.	06
(B)	Define the all step of selection sort as an ascending order of following data?	04
	15, 23, 17, 9, 20, 28, 25	
Q.6(A)	Write a detail note on processing of sequential file.	06
(B)	Write a short note on Multiple buffering.	04
	OR	
Q.6(A)	Write a detail note on processing of Direct file.	06
(B)	Write a short note on Single buffering.	04

